

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			814	NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE.
20										817	USE ONLY FOR TIE IN
20			MODULAR DBB	K03AS22	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV	0	0	821	TO BE USED FOR LINES SUBJECT TO AIV OR WHERE REQUIRED BY STRESS.
30		VAV	MODULAR DBB	K03AS22	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.
40			MODULAR DBB	M03AS22	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - CONTROLLED BORE TYPE 2BVR	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
50		VAW	MODULAR DBB	M03ES22	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
60		VAW	MODULAR DBB	M03ES22	1	1	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
70							***** BALL VALVES *****			FF5	FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES
70										FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
80											
80		VB2	BALL	A03AS21	3	4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO 718 BOLT	0	0		
90			BALL	A03AS22	3	4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL RB LO 718 BOLT	0	0		
100		VB2	BALL	A03AS27	6	8	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO 718 BOLT	0	0		
110			BALL	A03AS28	6	8	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION METAL RB GO 718 BOLT	0	0	SB6	USE JACKSCREWS FOR PADDLE
120		VB2	BALL	B03AS21	0.5	2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO 718 BOLT	0	0		BLANKS/SPACERS AND FLANGE SPREADERS
130		VB2	BALL	B03AS21	3	3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO 718 BOLT	0	0		FOR SPECTACLE BLINDS.
140			BALL	B03AS22	0.5	2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO 718 BOLT	0	0	UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST
150		VA5	BALL	B03AS22	3	3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO 718 BOLT	0	0		
160	U04	VA5	BALL	B03AS23	4	4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB GO 718 BOLT	0	0		
170	U04	VA5	BALL	B03AS24	4	4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB GO 718 BOLT	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL
180		VB2	BALL	B03AS27	6	8	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT	0	0		CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH
190		VB2	BALL	B03AS27	10	24	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT	0	0		CHEMICAL SEALS.
200		VA5	BALL	B03AS28	6	24	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO 718 BOLT	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
210							***** BUTTERFLY VALVES *****			VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
210										VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
220			BUTTERFLY	T03AS65	10	24	BUTTERFLY 300# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO 718 BOLT	0	0		
230							***** GLOBE VALVES *****				
230											
240											
240			GLOBE	L03AS55	0.75	1.5	GLOBE 300# RF SS A182 F316 TRIM 16 OS&Y HW 718 BOLT	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
250			GLOBE	L03AS56	2	6	GLOBE 300# RF SS A351 CF8M TRIM 16 OS&Y HW 718 BOLT	0	0		
260			GLOBE	L03AS57	8	10	GLOBE 300# RF SS A351 CF8M TRIM 16 OS&Y GO 718 BOLT	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH
270							***** CHECK VALVES *****				ENGINEERING APPROVAL.
270											
280											
280		VC1	CHECK	C03AS54	0.75	1.5	CHECK PISTON 300# RF SS A182 F316 TRIM 16 718 BOLT	0	0		
290	U04		CHECK	C03AS45	2	10	CHECK DUAL WAFER LUG 300# RF SS A351 CF8M TRIM 16 THROU BOLTED	0	0		
300			CHECK	C03AS52	12	24	CHECK DUAL DBL FLG 300# RF SS A351 CF8M TRIM 16	0	0		
310							***** PIPE *****				
310											
320											
320			PIPE	5365140	0.75	1.5	PIPE SCH 80S SMLS SS A312 TP316/316L PE	0	0		
330			PIPE	5365136	2	12	PIPE SCH 40S SMLS SS A312 TP316/316L BE	0	0		
340			PIPE	5365137	14	24	PIPE SCH 40 SMLS SS A312 TP316/316L BE	0	0		
350		814	PIPE	5365207	0.5	0.5	PIPE SCH 160 SMLS SS A312 TP316/316L PE	0	0		
360							***** FITTINGS *****				
360											
370											
370			ELBOW	5449033	0.75	1.5	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S	0	0		
380			ELBOW	5449138	2	12	ELL 90 DEG LR SCH 40S SS A403 WP316/316L-S	0	0		
390			ELBOW	5449032	14	24	ELL 90 DEG LR SCH 40 SS A403 WP316/316L-S	0	0		
400			ELBOW	5449146	0.75	1.5	ELL 45 DEG LR SCH 80S SS A403 WP316/316L-S	0	0		
410			ELBOW	5449144	2	12	ELL 45 SCH 40S SS A403 WP316/316L-S	0	0		
420			ELBOW	5449038	14	24	ELL 45 SCH 40 SS A403 WP316/316L-S	0	0		
430											
430			TEE	5449156	0.75	1.5	TEE SCH 80S SS A403 WP316/316L-S	0	0		
440			TEE	5449147	2	12	TEE SCH 40S SS A403 WP316/316L-S	0	0		
450			TEE	5449045	14	24	TEE SCH 40 SS A403 WP316/316L-S	0	0		
460			TEE RED	5448905	0.75	1	TEE RED SCH 80S SS A403 WP316/316L-S	1	1.5		
470			TEE RED	5449250	1.5	1.5	TEE RED SCH 40S/80S SS A403 WP316/316L-S	2	3		
480			TEE RED	5449184	2	10	TEE RED SCH 40S SS A403 WP316/316L-S	3	12		
490			TEE RED	5448897	6	12	TEE RED SCH 40/40S SS A403 WP316/316L-S	14	24		
500	U04	821	TEE RED	5449578	6	6	TEE RED SCH 120 SS A403 WP316/316L-S	10	10		
510	U04	821	TEE RED	5448906	6	6	TEE RED SCH 80/80S SS A403 WP316/316L-S	10	10		
520			TEE RED	5449046	14	20	TEE RED SCH 40 SS A403 WP316/316L-S	16	24		
530											
530			CAP	5449049	0.75	1.5	CAP SCH 80S SS A403 WP316/316L-S	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		All SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 300 RF TEMPERATURE RANGE: -120 / 180 °C	12/16/2024	BY OZAC	U04	Page 1 of 3	300P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
540			CAP	5449148	2	12	CAP SCH 40S SS A403 WP316/316L-S	0	0	
550			CAP	5449047	14	24	CAP SCH 40 SS A403 WP316/316L-S	0	0	
560										
560	U04		RED CONC	5449158	0.5	1	REDUCER CONC SCH 80S SS A403 WP316/316L-S	0.75	1.5	
570			RED CONC	5549110	0.75	1.5	REDUCER CONC SCH 40S/80S SS A403 WP316/316L-S	2	4	
580			RED CONC	5449149	2	10	REDUCER CONC SCH 40S SS A403 WP316/316L-S	3	12	
590			RED CONC	5448902	6	12	REDUCER CONC SCH 40/40S SS A403 WP316/316L-S	14	20	
600			RED CONC	5448903	14	20	REDUCER CONC SCH 40 SS A403 WP316/316L-S	16	24	
610			RED ECC	5449159	0.5	1	REDUCER ECC SCH 80S SS A403 WP316/316L-S	0.75	1.5	
620			RED ECC	5549310	0.75	1.5	REDUCER ECC SCH 40S/80S SS A403 WP316/316L-S	2	4	
630			RED ECC	5449150	2	10	REDUCER ECC SCH 40S SS A403 WP316/316L-S	3	12	
640			RED ECC	5449154	6	12	REDUCER ECC SCH 40/40S SS A403 WP316/316L-S	14	20	
650			RED ECC	5449155	14	20	REDUCER ECC SCH 40 SS A403 WP316/316L-S	16	24	
660										
660	U04	FF5	FLANGE	5504515	0.75	1.5	NIPOFLANGE FLAT BOTTOM 300# RF SCH 160 LTCS A350 LF2 CL.1	0	0	
670			OLETS	5504411	0.75	1.5	NIPOFLANGE 300# RF SCH 40S/80S SS A182 F316/316L	2	12	
680			OLETS	5504421	0.75	1.5	NIPOFLANGE 300# RF SCH 40/80S SS A182 F316/316L	14	24	
690			OLETS	5504032	2	2	NIPOFLANGE 300# RF SCH 40S/40S SS A182 F316/316L	2	12	
700			OLETS	5504033	2	2	NIPOFLANGE 300# RF SCH 40/40S SS A182 F316/316L	14	24	
710		UJ2	OLETS	5504132	1.5	1.5	NIPOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4	12	
720		UJ2	OLETS	5504142	1.5	1.5	NIPOFLANGE 300# RF SCH 40/160 SS A182 F316/316L	14	24	
730	U04	FF5	OLETS	5504520	0.75	1.5	NIPOFLANGE FLAT BOTTOM 300# RF SCH 80S SS A182 F316/316L	0	0	
740										
740			OLETS	5510354	0.75	1.5	WELDOLET SCH 40S/80S SS A182 F316/316L	2	12	
750			OLETS	5510364	0.75	1.5	WELDOLET SCH 40/80S SS A182 F316/316L	14	24	
760			OLETS	5509631	2	4	WELDOLET SCH 40S/40S SS A182 F316/316L	6	12	
770			OLETS	5509636	2	8	WELDOLET SCH 40/40S SS A182 F316/316L	14	24	
780										
780		UJ2	ELBOFLG	5516152	1.5	1.5	ELBOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4	12	
790		UJ2	ELBOFLG	5516158	1.5	1.5	ELBOFLANGE 300# RF SCH 40/160 SS A182 F316/316L	14	24	
800										
800			COUPLING	5480142	0.75	1.5	COUPLING 3000# SW SS A182 F316/316L	0	0	
810			TEE RED	5480134	0.75	1	COUPLING RED 3000# SW SS A182 F316/316L	1	1.5	
820										
820		UD7	FITTING SCRD	5498383	0.75	0.75	PLUG ROUND HEAD SCRD SS A182 F316/316L	0	0	
830										
830			OLETS	5537071	0.75	1.5	SOCKOLET 3000# SS A182 F316/316L	2	24	
840			OLETS	5533235	0.75	1.5	SOCKOLET FLAT BOTTOM 3000# SS A182 F316/316L	0	0	
850										
850		UD7	OLETS	5515771	0.75	0.75	THREDOLET 3000# SS A182 F316/316L	2	24	
860										
860							***** FLANGES *****			
870										
870			FLANGE	5851203	0.5	2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0	
880										
880	U04	821	FLANGE	5582991	6	6	FLG WN 300# RF SS A182 F316/316L SCH 80S BORE	0	0	
890			FLANGE	5582991	0.5	1.5	FLG WN 300# RF SS A182 F316/316L SCH 80S BORE	0	0	
900			FLANGE	5582995	2	12	FLG WN 300# RF SS A182 F316/316L SCH 40S BORE	0	0	
910			FLANGE	5582997	14	24	FLG WN 300# RF SS A182 F316/316L SCH 40 BORE	0	0	
920	U04		FLANGE	5582990	6	6	FLG WN 300# RF SS A182 F316/316L SCH 120 BORE	0	0	
930	U04	FS1	FLANGE	5583535	2	6	FLG WN 600# RF SS A182 F316/316L SCH 40S BORE	0	0	
940		FA8	TEE RED	5600656	0.5	0.5	FLG SCRD (FNPT) REDUCING 300# RF SS A182 F316/316L	0.75	1.5	
950			BLIND	5616529	0.5	24	FLG BLIND 300# RF SS A182 F316/316L	0	0	
960	U04		FLANGE	5583970	2	2	FLG WN 1500# RF SS A182 F316/316L SCH 40S BORE	0	0	
970	U04		FLANGE	5583979	1	1	FLG WN 1500# RF SS A182 F316/316L SCH 80S BORE	0	0	
980	U04		FLANGE	5583979	0.5	0.5	FLG WN 1500# RF SS A182 F316/316L SCH 80S BORE	0	0	
990	U04	817	FLANGE	5584157	2	2	FLG WN 1500# RTJ SS A182 F316/316L SCH 40S BORE	0	0	
1000	U04	FS1		5582998	0.5	0.5	FLG WN 600# RF SS A182 F316/316L SCH 80S BORE	0	0	
1010	U04	FS1	FLANGE	5582998	1	1.5	FLG WN 600# RF SS A182 F316/316L SCH 80S BORE	0	0	
1020	U04	817		5558034	3	3	FLG WN 600# RTJ SS A182 F316/316L SCH 40S BORE	0	0	
1030	U04		FLANGE	5583021	4	4	FLG WN 900# RF SS A182 F316/316L SCH 40S BORE	0	0	
1040										
1040							***** ORIFICE FLANGES *****			
1050										
1050			ORIFICE FLG	5640598	2	12	WN ORIF SET 300# RF SS A182 F316/316L SCH 40S BORE	0	0	
1060			ORIFICE FLG	5640599	14	24	WN ORIF SET 300# RF SS A182 F316/316L SCH 40 BORE	0	0	
1070										
1070							***** PIPE NIPPLES *****			
1080										
1080	U04	814					NIPPLE PBE CUT FROM PIPE			
1090		814		0			NIPPLE PBE CUT FROM PIPE			
1100							***** GASKETS *****			

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1110										
1110			GASKET	5673033	0.5	24	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1120			GASKET	5673103	0.5	6	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1130	U04		GASKET	5672251	0.5	0.5	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1140	U04		GASKET	5672251	1	1	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1150	U04		GASKET	5672251	2	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1160	U04	817	GASKET	5674479	2	2	GASKET 1500# RTJ SS OCTAGONAL RING	0	0	
1170	U04	817		5674490	3	3	GASKET 600# RTJ SS OCTAGONAL RING	0	0	
1180	U04		GASKET	5673117	4	4	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1190										
1190	B1						***** BOLTING *****			
1200										
1200			BOLTING	5675677	0.5	99	STUD-BOLT A320 GR B8M CLASS 2/A194 GR 8MA NUTS	0	0	
1210				5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1220			BOLTING	5675776	0.5	99	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
1230										
1230	SB6		BOLTING	5675676	12	24	JACK SCREW HEX HEAD A320 B8M - CLASS 2	0	0	
1240										
1240							***** BLINDS & SPACERS *****			
1250										
1250			SPEC BLIND	5750422	0.75	10	SPECTACLE BLIND 300# RF SS A240 GR 316	0	0	
1260										
1260			PADDLE	5750522	12	24	PADDLE BLANK 300# RF SS A240 GR 316	0	0	
1270			PADDLE	5750622	12	24	PADDLE SPACER 300# RF SS A240 GR 316	0	0	
1280										

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